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Garden blower

Abstract

The present disclosure provides a garden blower, including a housing and a duct assembly, the duct assembly includes an outer duct, a motor fan blade assembly and a guide cone; the outer duct and the rear casing are arranged in such a way that there is an annular interval therebetween, and a plurality of first anti-vibration components are arranged between the outer duct and the holding casing; the first anti-vibration components are close to the motor fan blade assembly, and form a first anti-vibration plane perpendicular to the motor axis of the motor fan blade assembly. This technical solution can suppress the transmission of vibration generated by the rotation of the motor fan blade assembly to other components, especially the holding casing, and effectively avoids the mutual influence between the vibration of the motor fan blade assembly and the disturbance of the rear casing caused by the airflow.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims priority to Chinese Patent Application No. 202410323062.X, filed on Mar. 21, 2024 before the China National Intellectual Property Administration, the disclosure of which is incorporated herein by reference in entirety.
TECHNICAL FIELD
(2) The present disclosure relates to a garden blower, belonging to the field of garden tools.
BACKGROUND
(3) The blower is a commonly used garden electric tool, mainly used to clean fallen leaves, grass clippings, road dust, etc. by airflow. Generally speaking, there is a gap between the outlet position of the duct and the housing of the existing blower. The airflow from the duct to the air outlet will flow out from the gap, resulting in poor sealing of the blower. In the prior art, in order to solve this problem, the duct is hard-connected to the housing at the outlet position. However, it will increase the vibration of the blower, causing noise and discomfort.
(4) At present, there is provided a Chinese invention patent application with application number CN202110276471.5, titled with blower, it sets a noise reduction seal between the housing and the duct to solve the above problem. However, the noise reduction and vibration reduction effect of the structure still has a lot of room for improvement, and when using the blower, you will still feel a lot of vibration and noise.
SUMMARY
(5) The present disclosure aims to provide a garden blower that can effectively prevent vibration.
(6) In view of this, the present disclosure provides a garden blower, comprising: a housing having an air inlet end and an air outlet end; and a duct assembly disposed in the housing and located between the air inlet end and the air outlet end, wherein the duct assembly comprises an outer duct, a motor fan blade assembly and a guide cone, and the motor fan blade assembly and the guide cone are disposed in the outer duct; wherein the outer duct and a rear casing of the housing are arranged in such a way that there is an annular interval therebetween, and the outer duct and the rear casing are deliberately set at a safe distance from each other so as to maintain an absolute non-contact state, forming a negative pressure inlet; wherein a first anti-vibration component is provided between the outer duct and a holding casing of the housing, so that the outer duct and the holding casing are spaced apart by the first anti-vibration component to maintain a state of non-direct contact; wherein the first anti-vibration component is close to the motor fan blade assembly and forms a first anti-vibration plane perpendicular to a motor axis of the motor fan blade assembly; the first anti-vibration plane is parallel to a direction of an exciting force of the vibration generated by the motor fan blade assembly, and passes through the motor fan blade assembly and/or a holding handle located above the motor fan blade assembly.
(7) According to some embodiments of the present disclosure, a second anti-vibration component is provided between one end of the outer duct away from the motor fan blade assembly and the holding casing, so that the outer duct and the holding casing are spaced apart by the second anti-vibration component to maintain a state of non-direct contact; wherein the second anti-vibration

component forms a second anti-vibration plane parallel to the first anti-vibration plane, and the guide cone is at least partially located between the first anti-vibration plane and the second anti-vibration plane.
(8) According to some embodiments of the present disclosure, the outer duct comprises a support end located at the air outlet, the support end is configured to support a cone end of the guide cone, the other end of the guide cone is connected to a motor seat, and the second anti-vibration component is sandwiched between an outer wall of the support end and an inner wall of the holding casing.
(9) According to some embodiments of the present disclosure, the second anti-vibration component is close to the guide cone, and the second anti-vibration plane formed by the second anti-vibration component passes through the guide cone and is perpendicular to an extension direction of an axis of the guide cone.
(10) According to some embodiments of the present disclosure, a center of gravity of the motor fan blade assembly is located between the first anti-vibration plane and the second anti-vibration plane.
(11) According to some embodiments of the present disclosure, a distance L**2** from the second anti-vibration plane to the center of gravity is greater than a distance L**1** from the first anti-vibration plane to the center of gravity.
(12) According to some embodiments of the present disclosure, under the rotation of the motor fan blade assembly, an airflow flows at high speed from the air inlet end to the air outlet end, and an accommodating chamber connected to the negative pressure inlet is provided between the rear casing and the holding casing, and a controller is provided in the accommodating chamber.
(13) According to some embodiments of the present disclosure, the rear casing is provided with an installation port, the controller is embedded in the installation port for positioning, heat dissipation fins of the controller are exposed to the installation port, and the airflow flowing at high speed from the air inlet end to the air outlet end is suitable for passing over the heat dissipation fins of the controller.
(14) According to some embodiments of the present disclosure, the outer duct comprises a motor fan blade bracket and a connecting pipe connected to each other, the motor fan blade bracket is provided with a cable hole for allowing a power line connecting the controller to a motor to pass through, and an outer wall of the motor fan blade bracket is formed with multiple reinforcing ring ribs, and the multiple reinforcing ring ribs cover a non-connected part of the motor fan blade bracket and the connecting pipe.
(15) Compared with the prior art, the garden blower provided by the present disclosure has the following beneficial technical effects:
(16) When the garden blower is in operation, the vibration source mainly comes from the vibration generated by the rotation of the motor fan blade assembly and the disturbance of the airflow to the housing due to turbulence. By arranging multiple first anti-vibration components between the outer duct and the holding casing, the vibration generated by the rotation of the motor fan blade assembly can be suppressed from being transmitted to other components, especially the holding casing. At the same time, the annular interval between the outer duct and the rear casing can effectively avoid the mutual influence between the rotation vibration of the motor fan blade assembly and the disturbance of the rear casing due to the airflow. At this time, whether it is the rotation vibration of the motor fan blade assembly or the disturbance of the rear casing due to the airflow, the two will not contact each other, and thereby no vibration superposition will be generated.
(17) According to another aspect of the present disclosure, there is provided another garden blower, comprising: a housing having an air inlet end and an air outlet end; and a main air path component, comprising a duct assembly, and forming a main air path suitable for airflow to flow from the air inlet end to the air outlet end at high speed, an accommodating chamber being formed between the main air path component and the housing, and an electronic device being provided in the accommodating chamber, wherein the duct assembly comprises an outer duct and a motor fan blade

assembly arranged in the outer duct; wherein the outer duct and a rear casing of the housing are arranged in such a way that there is an annular interval therebetween, and a negative pressure inlet connected to the accommodating chamber is formed between the outer duct and the rear casing; during operation of the garden blower, the heat emitted by the electronic device flows from the accommodating chamber into the main air path through the negative pressure inlet; wherein a first anti-vibration component is provided between the outer duct and a holding casing of the housing, so that the outer duct and the holding casing are spaced apart by the first anti-vibration component, and during operation of the garden blower, the outer duct and the holding casing are not in direct contact; wherein the first anti-vibration component is close to the motor fan blade assembly and forms a first anti-vibration plane perpendicular to a motor axis of the motor fan blade assembly; the first anti-vibration plane is parallel to a direction of an exciting force of the vibration generated by the motor fan blade assembly.
(18) According to some embodiments of the present disclosure, the electronic device comprises a controller.
(19) According to some embodiments of the present disclosure, the rear casing constitutes a part of the main air path component.
(20) This garden blower has the above-mentioned beneficial technical effects, same to the above garden blower, which will not be described repeatedly here. In addition, the heat generated by the electronic device in the accommodating chamber can be introduced into the main air path through the formed negative pressure inlet, so as to effectively dissipate the heat of the electronic device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In order to more clearly illustrate the specific implementation of the present disclosure or the technical solution in the prior art, the drawings required for the description of the specific implementation or prior art will be briefly introduced below. Obviously, the drawings in the following description are only some implementations of the present disclosure. For ordinary technicians in this field, other drawings may also be obtained based on these drawings without creative work.
(2) FIG. **1** is a schematic structural view of the first specific embodiment of the present disclosure;
(3) FIG. **2** is a structural cross-sectional view of the first specific embodiment of the present disclosure;
(4) FIG. **3** is a partial enlarged schematic view of portion A of the present disclosure in FIG. **2**;
(5) FIG. **4** is a partial enlarged schematic view of portion B of the present disclosure in FIG. **2**;
(6) FIG. **5** is a schematic explosive view of the possible embodiment 1 of the present disclosure showing the second anti-vibration plane S**2**;
(7) FIG. **6** is a schematic view showing anti-vibration plane relationship of the possible embodiment 1 of the present disclosure;
(8) FIG. **7** is a schematic explosive view of the possible embodiment 2 of the present disclosure showing the second anti-vibration plane S**2**;
(9) FIG. **8** is a schematic view showing anti-vibration plane relationship of the possible embodiment 2 of the present disclosure;
(10) FIG. **9** is a schematic view of the heat dissipation structure of the controller in the second specific embodiment of the present disclosure;
(11) FIG. **10** is a partial enlarged schematic view of portion C of the present disclosure in FIG. **9**;
(12) FIG. **11** is a schematic structural view of the outer duct of the present disclosure.
(13) Garden blower **100**; housing **10**; air inlet end **101**; air outlet end **102**; rear casing **10***a*; holding casing **10***b*; holding handle **10***b***1**; duct assembly **20**; outer duct **201**; motor fan blade assembly **202**;

guide cone **203**; first anti-vibration component **30**; first anti-vibration plane S**1**; second anti-vibration component **40**; second anti-vibration plane S**2**; support end **60**; center of gravity G; accommodating chamber **70**; controller **80**; installation port **10***a***1**; motor fan blade bracket **201***a*; connecting pipe **201***b*; first connecting part **201***a***1**; second connecting part **201***b***1**; cable hole S; non-connected part **201***a***2**; main air path component **200**; main air path **300**; motor axis L.

DETAILED DESCRIPTION OF EMBODIMENTS

(14) The technical solution of the present disclosure will be clearly and completely described below in conjunction with the accompanying drawings. Obviously, the described embodiments are part of the present disclosure, but not all of them. Based on the embodiments of the present disclosure, all other embodiments obtained by ordinary technicians in this field without creative work fall within the scope of protection of the present disclosure.

First Embodiment

(15) As shown in FIGS. **1** and **2**, a garden blower **100** includes a housing **10** and a duct assembly **20**.

(16) The housing **10** has an air inlet end **101** and an air outlet end **102**, wherein the housing **10** is composed of at least a rear casing **10***a* and a holding casing **10***b*, the rear end of the rear casing **10***a* forms the air inlet end **101**, and the holding casing **10***b* has a holding handle **10***b***1** for the user to hold and use.

(17) The duct assembly **20** is arranged in the housing **10**, and the duct assembly **20** is located between the air inlet end **101** and the air outlet end **102**.

(18) The duct assembly **20** comprises an outer duct **201**, a motor fan blade assembly **202** and a guide cone **203**, the motor fan blade assembly **202** and the guide cone **203** are arranged in the outer duct **201**; the user holds the holding handle **10***b***1** and actuates the trigger switch, the garden blower **100** receives power and starts to work, the fan blades of the motor fan blade assembly **202** start to rotate under the drive of the motor, and under the rotation of the motor fan blade assembly **202**, the airflow flows from the air inlet end **101** through the outer duct **201** and flows to the air outlet end **102** at a high speed.

(19) Further, referring to the partial enlarged view shown in FIG. **3** and FIG. **5**, the outer duct **201** and the rear casing **10***a* of the housing **10** are arranged with an annular spacing/interval. Preferably, the annular spacing/interval is 1 mm-2 mm, and more preferably, the annular spacing/interval is set to 1.5 mm. For example, the annular diameter of the rear casing **10***a* is 99.4 mm, and the annular diameter of the corresponding part of the outer duct **201** is 96.4 mm. Thus, the annular spacing/interval between the two is just 1.5 mm. In this way, the outer duct **201** and the rear casing **10***a* are deliberately set outside the safe distance of non-contact, maintaining an absolute non-contact state.

(20) This setting can effectively avoid the mutual influence between the vibration of the motor fan blade assembly **202** and the disturbance caused by the airflow of the rear casing **10***a*. When the garden blower **100** is in operation, the vibration source mainly comes from the vibration generated by the rotation of the motor fan blade assembly **202** and the disturbance of the airflow to the housing **10** due to turbulence. When the annular interval between the outer duct **201** and the rear casing **10***a* of the housing **10** maintains an absolute non-contact state, whether it is the rotation vibration of the motor fan blade assembly **202** or the disturbance of the rear casing **10***a* due to the airflow, the two will not touch each other. The two are independent of each other and are not affected by each other, and no vibration superposition will be generated.

(21) At the same time, a first anti-vibration component **30** is provided between the outer duct **201** and the holding casing **10***b* of the housing **10**; the first anti-vibration component **30** allows the outer duct **201** and the holding casing **10***b* to be spaced apart by the anti-vibration component to maintain a non-direct contact state. The first anti-vibration component **30** is preferably made of elastic material, such as rubber, sponge, etc. Thus, the outer duct **201** and the holding casing **10***b* are in indirect contact with each other through the anti-vibration component. Through the elastic

deformation of the first anti-vibration component **30**, the vibration of the outer duct **201** generated by the rotation of the motor can be suppressed from being transmitted to the holding casing **10***b*. Herein, the first anti-vibration components **30** are preferably multiple, and the multiple first anti-vibration components **30** are arranged annularly. Of course, the first anti-vibration component **30** may be one. If so, the first anti-vibration component **30** is an elastic component that is annular as a whole.
(22) Specifically, when the first anti-vibration components **30** are preferably multiple, the outer duct **201** is provided with protrusions protruding toward the holding casing **10***b*, the first anti-vibration components **30** are installed on the protrusions, and the holding casing **10***b* is correspondingly provided with receiving portions, so that the first anti-vibration components **30** are clamped between the outer duct **201** and the holding casing **10***b*, specifically, between the protrusions of the outer duct **201** and the receiving portions of the holding casing **10***b*.
(23) Referring to FIGS. **5** and **6**, the first anti-vibration components **30** are close to the motor fan blade assembly **202** and form a first anti-vibration plane S**1** which is perpendicular to the motor axis L of the motor fan blade assembly **202**. The first anti-vibration plane S**1** is parallel to the direction of the exciting force of the vibration generated by the motor fan blade assembly **202** and passes through the motor fan blade assembly **202**. At this time, the first anti-vibration plane S**1** is close to the center of the vibration source (motor fan blade assembly **202**), which can largely offset the exciting force of the vibration generated by the motor fan blade assembly **202**, and can more effectively suppress the vibration generated by the motor fan blade assembly **202**.
(24) Alternatively, the first anti-vibration plane S**1** passes through the holding handle **10***b***1** located above the motor fan blade assembly **202**. Since it is well known that there is a holding/gripping space suitable for fingers to be inserted below the holding/gripping handle **10***b***1**, that is, the part of the holding handle **10***b***1** for users to grip is separated and spaced from the holding casing **10***b* below it, which can play a role in reducing vibration transmission. When the first anti-vibration plane S**1** passes through the holding handle **10***b***1**, the holding handle **10***b***1**, as the part of the user holding the housing, has a small vibration feeling.
(25) Or further, it passes through the motor fan blade assembly **202** and the holding handle **10***b***1** located above the motor fan blade assembly **202** at the same time. Thus, on the one hand, the first anti-vibration plane S**1** is close to the center of the vibration source (motor fan blade assembly **202**), which can largely offset the exciting force of the vibration generated by the motor fan blade assembly **202**, and can more effectively suppress the vibration generated by the motor fan blade assembly **202**, on the other hand, the vibration transmission from the holding casing **10***b* to the holding portion of the holding handle **10***b***1** is weakened through the holding space below the holding handle **10***b***1**.
(26) Further, referring to FIGS. **5** and **6**, as well as FIGS. **7** and **8**, a second anti-vibration component/second anti-vibration components **40** are provided between the end of the duct **201** away from the motor fan blade assembly **202** and the holding casing **10***b*. Similarly, the second anti-vibration component **40** makes the duct **201** and the holding casing **10***b* still separated by the anti-vibration component, and still maintains a state of non-direct contact. In this way, the second anti-vibration component **40** forms a second anti-vibration plane S**2** parallel to the first anti-vibration plane S**1**, and the guide cone **203** is at least partially located between the first anti-vibration plane S**1** and the second anti-vibration plane S**2**. Similarly, the second anti-vibration component **40** is preferably made of elastic material.
(27) Those skilled in the art know that the guide cone **203** is connected to a motor seat for installing the motor. When the motor fan blade assembly **202** rotates, it will inevitably cause the guide cone **203** to vibrate. By setting the second anti-vibration plane S**2**, the vibration of the guide cone **203** can be suppressed. Specifically, the arrangement of the second anti-vibration plane S**2** has multiple possible implementation methods.
(28) [Possible Embodiment 1 of the Second Anti-Vibration Plane S**2**]

(29) In one possible embodiment, referring to FIGS. **2** and **4**, and in combination with FIGS. **5** and **6**, the outer duct **201** includes a support end **60** located at the air outlet, and the support end **60** is configured to support the cone end of the guide cone **203**, and the other end of the guide cone **203** is connected to the motor seat. The second anti-vibration component **40** is sandwiched between the outer wall of the support end **60** and the inner wall of the holding casing **10***b*.
(30) Herein, both ends of the guide cone **203** are positioned. On the one hand, the second anti-vibration plane S**2** can effectively prevent vibration of the guide cone, and combined with the first anti-vibration plane S**1**, it can further effectively prevent vibration, on the other hand, the positioning of the two ends also has a counteracting effect on the disturbance generated by the high-speed airflow.
(31) [Possible Embodiment 2 of the Second Anti-Vibration Plane S**2**]
(32) In another embodiment, as shown in FIGS. **7** and **8**, the second anti-vibration components **40** are close to the guide cone **203**, and the second anti-vibration plane S**2** formed by them passes through the guide cone **203** and is perpendicular to the axis extension direction of the guide cone **203**.
(33) [Possible Embodiment 3 of the Second Anti-Vibration Plane S**2**]
(34) In another embodiment, multiple second anti-vibration planes can be allowed, for example, the second anti-vibration components **40** are sandwiched between the outer wall of the support end **60** and the inner wall of the holding casing **10***b* to form one second anti-vibration plane, and at the same time, other second anti-vibration components **40** are close to the guide cone **203**, and another second anti-vibration plane formed by them passes through the guide cone **203**. In this way, the effect of suppressing the vibration of the guide cone **203** will be better.
(35) Further, the center of gravity G of the motor fan blade assembly **202** is located between the first anti-vibration plane S**1** and the second anti-vibration plane S**2**. The center of the vibration source is located between the two anti-vibration planes, which can further suppress the vibration.
(36) Further, the distance L**2** from the second anti-vibration plane S**2** to the center of gravity G is greater than the distance L**1** from the first anti-vibration plane S**1** to the center of gravity G. In this way, the center of the vibration source will be closer to the first anti-vibration plane S**1**, and the first anti-vibration plane S**1** passes through the motor fan blade assembly **202** and/or the holding handle **10***b***1** located above the motor fan blade assembly **202**. Thus, on the one hand, the first anti-vibration plane S**1** is close to the center of the vibration source (motor fan blade assembly **202**), which can largely offset the exciting force of the vibration generated by the motor fan blade assembly **202**, and can more effectively suppress the vibration generated by the motor fan blade assembly **202**, on the other hand, through the holding space below the holding handle **10***b***1**, the vibration transmission from the holding casing **10***b* to the holding portion of the holding handle **10***b***1** is weakened.
(37) Further, as shown in FIG. **9**, under the rotation of the motor fan blade assembly **202**, the airflow flows at high speed from the air inlet end **101** to the air outlet end **102**, and a negative pressure inlet is formed at the annular interval between the outer duct **201** and the rear casing **10***a*. An accommodating chamber **70** connected to the negative pressure inlet is provided between the rear casing **10***a* and the holding casing **10***b*, and a controller **80** is provided in the accommodating chamber **70**. The heat generated by the controller **80** during operation needs to be dissipated.
(38) In the first embodiment, referring to the airflow arrows shown in FIG. **9**, when the garden blower **100** is in operation, the heat generated by the controller **80** flows into the main air path through the negative pressure inlet from the accommodating chamber **70** and is taken away, which can play a role in cooling the controller **80**.
(39) Further, the rear casing **10***a* is provided with an installation port **10***a***1**, the controller **80** is embedded in the installation port **10***a***1** for positioning, the heat dissipation fins of the controller **80** are exposed to the installation port **10***a***1**, and the airflow flows at a high speed from the air inlet end **101** to the air outlet end **102**, which is suitable for passing over the heat dissipation fins of the

controller **80**.
(40) At this time, the high-speed airflow of the main air path passes over the surface of the heat dissipation fins exposed to the installation port **10***a***1**, and further cools the controller **80**.
(41) In addition, as shown in FIG. **11**, the outer duct **201** includes a motor fan blade bracket **201***a* and a connecting pipe **201***b* connected to each other, the motor fan blade bracket **201***a* adopts a first connecting portion **201***a***1** to be connected to a corresponding second connecting portion **201***b***1** of the connecting pipe **201***b*, for example, a lock connection is adopted, the motor fan blade bracket **201***a* includes the above-mentioned motor seat, the motor fan blade bracket **201***a* is provided with a cable hole S for allowing the power line connecting the controller **80** with the motor to pass through, and the outer wall of the motor fan blade bracket **201***a* is formed with multiple reinforcing ring ribs, and the multiple reinforcing ring ribs cover the non-connected part **201***a***2** of the motor fan blade bracket **201***a* and the connecting pipe **201***b*.
(42) In this way, it can avoid the overall structural strength of the motor fan blade bracket **201***a* from being reduced due to the cable hole S, and avoid the motor fan blade bracket **201***a* from deformation due to long-term accommodation of the motor fan blade assembly as a vibration source.
(43) In short, when the garden blower **100** of the present disclosure is in operation, the vibration source mainly comes from the vibration generated by the rotation of the motor fan blade assembly **202** and the disturbance of the airflow to the housing **10** due to turbulence. By setting a plurality of first anti-vibration components **30** between the outer duct **201** and the holding casing **10***b*, the vibration generated by the rotation of the motor fan blade assembly **202** can be suppressed from being transmitted to other components, especially the holding casing **10***b*. At the same time, the annular interval between the outer duct **201** and the rear casing **10***a* can effectively avoid the mutual influence between the vibration of the motor fan blade assembly **202** and the disturbance of the rear casing **10***a* due to the airflow. The technical problem of the vibration of the blower in the prior art is effectively solved.
Second Embodiment
(44) Continuing to refer to FIGS. **9** and **10**, in the Second Embodiment, the garden blower **100** includes a housing **10** having an air inlet end **101** and an air outlet end **102**, and a main air path component **200**.
(45) The main air path component **200** includes a duct assembly **20**, and forms a main air path **300** suitable for high-speed flow of air from the air inlet end to the air outlet end. An accommodating chamber **70** is formed between the main air path component **200** and the housing **10**, and an electronic device is provided in the accommodating chamber **70**. The electronic device generates heat during operation, and it is necessary to perform heat dissipation treatment.
(46) The duct assembly **20** includes an outer duct **201** and a motor fan blade assembly **202** arranged in the outer duct **201**.
(47) The outer duct **201** of the duct assembly **20** is arranged in an annular interval with the rear casing **10***a* of the housing **10**, so that a negative pressure inlet connected to the accommodating chamber **70** is formed between the outer duct **201** and the rear casing **10***a*. When the garden blower **100** is in operation, the airflow flows in from the air inlet end **101** and flows at a high speed toward the air outlet end **102**, forming a negative pressure at the negative pressure inlet. The heat emitted by the electronic device in the accommodating chamber **70** flows from the accommodating chamber **70** into the main air path **300** through the negative pressure inlet. In this way, the heat generated by the electronic device when working will be immediately taken away by the high-speed airflow at the main air path **300**, achieving the technical effect of immediate heat dissipation and cooling of the electronic device in the accommodating chamber **70**. At the same time, the outer duct **201** and the rear casing **10***a* of the housing **10** are arranged in an annular interval. In order to ensure the existence of a stable negative pressure inlet, the outer duct **201** and the rear casing **10***a* maintain an absolute non-contact state. Whether the motor fan blade assembly **202** rotates and

vibrates, or the rear casing **10***a* is disturbed by the airflow, the two will not touch each other. The two are independent of each other and are not affected by each other, and no vibration superposition will be generated.
(48) In addition, a first anti-vibration component **30** is provided between the outer duct **201** of the duct assembly **20** and the holding casing **10***b* of the housing **10**, so that the outer duct **201** and the holding casing **10***b* are spaced apart by the first anti-vibration component **30**, and when the garden blower **100** is in operation, the outer duct **201** and the holding casing **10***b* are not in direct contact.
(49) The first anti-vibration component **30** is close to the motor fan blade assembly **202**, and forms a first anti-vibration plane S**1** perpendicular to the motor axis L of the motor fan blade assembly **202**, and the first anti-vibration plane S**1** is parallel to the direction of the exciting force of the vibration generated by the motor fan blade assembly **202**.
(50) As described above, the first anti-vibration component **30** enables the outer duct **201** and the holding casing **10***b* to be spaced apart by the anti-vibration component, so as to maintain a state of non-direct contact, and the first anti-vibration component **30** is preferably made of elastic material, such as rubber. Thus, the outer duct **201** and the holding casing **10***b* are in indirect contact with each other through the anti-vibration component, and the elastic deformation of the first anti-vibration component **30** can suppress the vibration of the outer duct **201** caused by the rotation of the motor from being transmitted to the holding casing **10***b*.
(51) In this way, the garden blower **100** can not only effectively prevent vibration, but also introduce the heat generated by the electronic device in the accommodating chamber into the main air path through the formed negative pressure inlet, and effectively dissipate the heat of the electronic device.
(52) More preferably, the electronic device includes a controller **80**, especially a brushless controller. In addition, still referring to FIG. **9**, the rear casing **10***a* constitutes a part of the main air path component **200**. That is, the main air path component **200** is at least partially composed of the duct assembly **20** and the rear casing **10***a*, and under the rotation of the motor fan blades, the main air path **300** is formed inside it. More preferably, the controller **80** is arranged between the rear casing **10***a* and the holding casing **10***b*.
(53) Finally, it should be noted that the above embodiments are only used to illustrate the technical solutions of the present disclosure, rather than to limit it. Although the present disclosure has been described in detail with reference to the above embodiments, it should be understood by those skilled in the art that they can still modify the technical solutions described in the above embodiments, or replace some or all of the technical features therein, and these modifications or replacements do not make the essence of the corresponding technical solutions deviate from the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. A garden blower, comprising: a housing having an air inlet end and an air outlet end; and a duct assembly disposed in the housing and located between the air inlet end and the air outlet end, wherein the duct assembly comprises an outer duct, a motor fan blade assembly and a guide cone, and the motor fan blade assembly and the guide cone are disposed in the outer duct; wherein the outer duct and a rear casing of the housing are arranged in such a way that there is an annular interval therebetween, and the outer duct and the rear casing are deliberately set at a safe distance from each other so as to maintain an absolute non-contact state, forming a negative pressure inlet, thereby whether it is rotation vibration of the motor fan blade assembly or disturbance of the rear casing due to the airflow, the outer duct and the rear casing would not contact each other, and thereby no vibration superposition would be generated; wherein a first anti-vibration component is provided between the outer duct and a holding casing of the housing, so that the outer duct and the holding casing are spaced apart by the first anti-vibration component to maintain a state of non-

direct contact; wherein the first anti-vibration component is close to the motor fan blade assembly and is situated on a first anti-vibration plane perpendicular to a motor axis of the motor fan blade assembly and parallel to a direction of an exciting force of the vibration generated by the motor fan blade assembly, and wherein the first anti-vibration plane passes through the motor fan blade assembly and/or a holding handle located above the motor fan blade assembly; wherein a second anti-vibration component is provided between one end of the outer duct, away from the motor fan blade assembly and the holding casing, so that the outer duct and the holding casing are spaced apart by the second anti-vibration component to maintain a state of non-direct contact; wherein the second anti-vibration component is situated on a second anti-vibration plane parallel to and non-coincident with the first anti-vibration plane, and wherein the guide cone is at least partially located between the first anti-vibration plane and the second anti-vibration plane; wherein under the rotation of the motor fan blade assembly, air is capable of flowing at high speed from the air inlet end to the air outlet end, and wherein an accommodating chamber connected to the negative pressure inlet is provided between the rear casing and the holding casing, and a controller is provided in the accommodating chamber; wherein the rear casing is provided with an installation port, and the controller is embedded in the installation port for positioning; wherein the outer duct comprises a support end located at the air outlet, the support end is configured to support a cone end of the guide cone, the other end of the guide cone is connected to a motor seat, and the second anti-vibration component is sandwiched between an outer wall of the support end and an inner wall of the holding casing.
2. The garden blower according to claim 1, wherein the second anti-vibration component is arranged relative to the guide cone, such that the second anti-vibration plane passes through the guide cone and is perpendicular to an extension direction of an axis of the guide cone.
3. The garden blower according to claim 1, wherein a center of gravity of the motor fan blade assembly is located between the first anti-vibration plane and the second anti-vibration plane.
4. The garden blower according to claim 3, wherein a distance L**2** from the second anti-vibration plane to the center of gravity is greater than a distance L**1** from the first anti-vibration plane to the center of gravity.
5. The garden blower according to claim 1, wherein heat dissipation fins of the controller are exposed to the installation port, and the airflow flowing at high speed from the air inlet end to the air outlet end is suitable for passing over the heat dissipation fins of the controller.
6. The garden blower according to claim 1, wherein the outer duct comprises a motor fan blade bracket and a connecting pipe connected to each other, the motor fan blade bracket is provided with a cable hole for allowing a power line connecting the controller to a motor to pass through, and an outer wall of the motor fan blade bracket is formed with multiple reinforcing ring ribs, and the multiple reinforcing ring ribs cover a non-connected part of the motor fan blade bracket and the connecting pipe.
7. A garden blower, comprising: a housing having an air inlet end and an air outlet end; and a main air path component, comprising a duct assembly, and forming a main air path suitable for airflow to flow from the air inlet end to the air outlet end at high speed, an accommodating chamber being formed between the main air path component and the housing, and an electronic device being provided in the accommodating chamber, and the electronic device comprising a controller, wherein the duct assembly comprises an outer duct and a motor fan blade assembly arranged in the outer duct; wherein the outer duct and a rear casing of the housing are arranged in such a way that there is an annular interval therebetween, and the outer duct and the rear casing are deliberately set at a safe distance from each other so as to maintain an absolute non-contact state, forming a negative pressure inlet connected to the accommodating chamber between the outer duct and the rear casing, thereby whether it is rotation vibration of the motor fan blade assembly or disturbance of the rear casing due to the airflow, the outer duct and the rear casing would not contact each other, and thereby no vibration superposition would be generated; wherein the rear casing is

provided with an installation port, the controller is embedded in the installation port for positioning; wherein during operation of the garden blower, the heat emitted by the electronic device flows from the accommodating chamber into the main air path through the negative pressure inlet; wherein a first anti-vibration component is provided between the outer duct and a holding casing of the housing, so that the outer duct and the holding casing are spaced apart by the first anti-vibration component, and during operation of the garden blower, the outer duct and the holding casing are not in direct contact; wherein the first anti-vibration component is close to the motor fan blade assembly and is situated on a first anti-vibration plane perpendicular to a motor axis of the motor fan blade assembly and parallel to a direction of an exciting force of the vibration generated by the motor fan blade assembly; wherein a second anti-vibration component is provided between one end of the outer duct, away from the motor fan blade assembly and the holding casing, so that the outer duct and the holding casing are spaced apart by the second anti-vibration component to maintain a state of non-direct contact; wherein the second anti-vibration component is situated on a second anti-vibration plane parallel to and non-coincident with the first anti-vibration plane, and wherein the guide cone is at least partially located between the first anti-vibration plane and the second anti-vibration plane; wherein the outer duct comprises a support end located at the air outlet, the support end is configured to support a cone end of the guide cone, the other end of the guide cone is connected to a motor seat, and the second anti-vibration component is sandwiched between an outer wall of the support end and an inner wall of the holding casing.

8. The garden blower according to claim 7, wherein the rear casing forms an outer boundary of the main air path component, thereby creating constrained air paths within.